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Wildfire Detection Using Satellite Imagery and Synthetic Thermal Images

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Abstract: A deep learning-based technique of fusion between RGB and synthetically created thermal images for wildfire detection is presented in this work. Binary fire discrimination is obtained using a four-channel CNN model based on ResNet-34 architecture by fusing RGB inputs with synthetically created thermal maps created through color–intensity conversions to mimic infrared cues without physical thermal sensors. Fusion enhances discrimination of features under varying illumination, smoke, and occlusion. Experiments on a hybrid real–synthetic dataset prove that the model has attained an accuracy of over 95% and fewer false alarms compared to RGB-only CNNs. With minimal computational cost and high reliability, the system is appropriate for real-time wildfire detection on UAVs and edge devices.

Keywords—Wildfire detection, image fusion, simulated thermal imaging, ResNet-34, deep learning, edge computing.

I. INTRODUCTION

Wildfires have gained prominence as a major global concern, causing the destruction of millions of hectares of forests every year and putting at risk the lives of humans, the survival of different species, and the infrastructures. Among the causes of increased wildfires, the most dominant ones are climate change, deforestation, and the activities of humans [1], [2]. Early detection is paramount in saving the environment and the economy; however, the traditional methods that are used for this purpose like manual observation, satellite watches, and ground sensors have the drawbacks of being slow, having low spatial resolution, and being affected by the weather [3]

Recent technological progress in computer vision and deep learning have enabled the use of image-based, automated fire detection systems that are quicker and more accurate compared to conventional techniques [4]. Specifically, Convolutional Neural Networks (CNNs) have been found capable of determining fire regions at high accuracy through processing visual information related to texture, color, and movement [5]. Despite the above enhancements, RGB imagery remains the only foundation for the majority of contemporary systems, rendering it particularly vulnerable to lighting changes, smoke density, and physical obstructions. TIR images, by contrast, convey information according to thermal emissions that are unaffected by the availability of light, therefore serving as an excellent complementary modality for dependable fire detection that is likewise very informative [6]. Yet, thermal data collection typically involves expensive dual-camera setups that limit field deployment and scalability [7].

To overcome such constraints, this research in this paper envisions a four-channel CNN model that merges RGB and synthetic thermal data into an integrated deep model. For the proposed model, it builds upon the ResNet-34 backbone to handle one supplementary thermal channel from RGB images using color–intensity mapping. This complementary approach extracts visual as well as heat signals, thus improving the model's inference of fire across different environmental scenarios. The architecture is formulated for real-time inference and can be easily deployed on unmanned aerial vehicles (UAVs) and edge computing platforms.

The performance of the proposed method is verified using FlameVision dataset containing real RGB images of fires and artificially produced thermal versions from multimodal datasets such as SWIFT [8], FLAME2 [9], and RGBT-3M [10]. Experimental findings reveal that 4-channel model has over 95% classification accuracy and outperforms both baseline RGB-only CNNs and competitive architectures such as EfficientNet-B0. This research highlights the possibility of using simulated thermal fusion as a low-cost, scalable solution for the early detection and monitoring of wildfires.

II. RELATED WORK

A. Traditional Image-Based Fire Detection

Early detection approaches of wildfires were mainly carried out through colour space thresholding and/or motion-based

analysis for fire-like region segmentation in visual imagery [11]. The YUV and HSV color space based methods were computationally low demanding and feasible for real time implementations, but they were very susceptible to variations caused by different factors (e.g., reflection, variation in light intensity conditions and background clutter) depending on the static coloured heuristic used [12].

One of the earliest well-distinguished investigations, Image Processing-Based Forest Fire Detection [13], used pixel-wise color ratio analysis, flame contour tracing and temporal variation modelling to distinguish fire from non-fire zones. Although these classical image-processing based methods established the foundation for vision-based fire detection, they were mostly not robust enough especially in complex outdoor scenes and did not have good scalability. Moreover, their hand-crafted features and threshold parameters made them not well adapted to fire detection in different scenarios, which much constrained the practical application of these two detection methods in autonomous monitoring systems.

B. Deep Learning for Fire Recognition

Deep learning revolutionized the fire detection task with end-to-end discriminative feature learning from data. In particular, CNNs showed great advantage in modelling complicated spatial and spectral patterns over some hand-crafted features. Ghali and Akhloufi [8] proposed the CT-Fire and DC-Fire models which used DenseNet-201, EfficientNet-B5, and RegNetY-16GF networks to process RGB–Thermal aerial isometric views that obtained high classification accuracy on FLAME2 dataset. Also, Zhang et al. [9] proposed the aforementioned CP-YOLOv11-MF detector, in which the UAV-captured RGB–Thermal data is modally fused and could contribute to 96.3 % mAP@50 on RGBT-3M dataset. These results justify the effectiveness of multimodal fusion in boosting detection robustness with poor visual conditions.

Fernando et al. [10] established a large-scale synthetic wildfire dataset: SWIFT, which consists of RGB, smoke, and thermal-inspired rendered images. Their results demonstrated that sim-to-real transfer learning enables CNN-based detectors trained on synthetic data to generalize well in real-world settings, and more generally benefit from simulated modalities when access is limited. And, articles such as those of the CSEE'22 World Congress [14] in ICEPTP 204 were focusing on the role of AI-enabled multimodal sensing for environmental hazard management (with hybrid fusion seen as pivotal) in early warning and response.

C. Limitations of the Existing Approaches

Nevertheless, these recent advances on real-time wildfire detection suffer from inherent restrictions which limit their applicability in practical and unequivocal manner. The state of the art image sensing solutions are now dominated by several cost-, calibration- and energy-consuming dual-camera RGB–Thermal systems. Vision only models still succumb to environmental distortions such as smoke, haze, or changes in lighting conditions. In the addition, most deep models trained on high-resolution data are usually large and demand huge computational resources, making them too heavy for real-time flight of UAVs or edge devices. For this work, a 4 model ResNet-34 architecture is adopted in which RGB and simulated thermal imaging by color intensity mapping are combined. This approach maintains the strength of thermal fusion in a multi-modal sense, and does not rely on dedicated hardware, while this has been able to achieve a good tradeoff between detection accuracy, computational efficiency and deployment (resource limitation).

III. DATASET AND PREPROCESSING

A. FlameVision Dataset Overview

The FlameVision dataset was built with the main purpose of facilitating RGB-thermal fusion method evaluations for fire detection. Its conceptualization is theoretically inspired by already available multimodal datasets, i.e., FLAME2, RGBT-3M, and SWIFT, in order to facilitate experimentation at low costs by using simulated thermal data rather than hardware-captured infrared imagery [8], [9], [10].

The data set includes 12,000 images captured under diverse environmental conditions, i.e., daylight intensity, weather, smoke concentration, and texture of background. This is done to make the model resilient to real-world visual variability. In the interest of reproducible experimentation, this paper divides the dataset into training, validation, and test subsets in the proportion of 70%, 15%, and 15% via stratified sampling that preserves class balance across splits.

Each image is a four-channel tensor where the first three channels are the visible spectrum and the fourth channel is simulated thermal intensity.
This second channel enhances the learning of the model from combined visual and radiometric data and improves the discriminability of flame regions from background clutter-like sun reflection or highly colored objects. By bringing together spatial and heat-related information, FlameVision bridges the gap between low-cost RGB inputs and denser semantic information typically obtained from infrared sensors.

B. Thermal Simulation Process

Thermal infrared cameras are required heat signatures for active wildfire detection but are still too costly and power intensive to deploy extensively. FlameVision thus relies on a simulation-based synthetic thermal generation process, driven by simulation approaches developed in SWIFT [10].

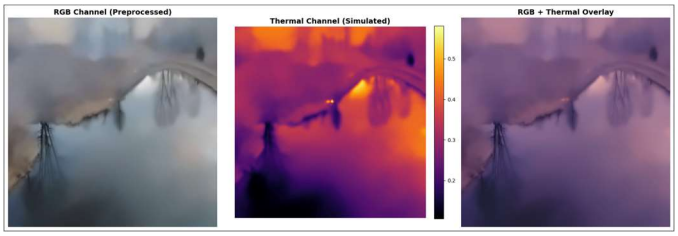


Fig 1. Comparison between RGB image, simulated thermal map, and overlay fusion result.

Each RGB image is transformed into the HSV color space to decouple chromatic tone or hue from intensity or value and from chroma or saturation. Pixels of warm hues—typically reds, oranges, and yellows ($H \in [0°, 60°]$)—are selected as the regions that could possibly contain a flame. A heat-intensity mapping function mixes brightness with saturation for these pixels to estimate the energetic signature of flames and produce a grayscale heat map where larger pixel values reflect higher simulated thermal intensity.

The generated heat map is normalized and appended to the RGB channels as the fourth channel, resulting in a fused representation that is structurally very similar to real RGB–thermal imagery but remains computationally lightweight. The synthetic thermal channel captures salient physical features such as localized brightness gradients, color–temperature correlation, and flame morphology that closely mimic patterns in true thermal infrared images. A conceptual comparison between generated and real thermal images is provided in Fig. 1.

This simulation strategy allows training multimodal deep models without requiring special sensors, which lowers the barrier to conducting research and rapid prototyping in resource-constrained environments.

C. Data Augmentation

To improve model generalization and reduce overfitting, FlameVision employs a comprehensive data augmentation pipeline during training. The augmentation strategy follows standard, empirically validated practices in image-based modeling and wildfire detection literature [18].

The core transformations include random rotations ±30° for the simulation of viewpoint changes, horizontal and vertical flips for mirrored viewpoints, and color jitter to emulate illumination and atmospheric scattering variations. CoarseDropout occludes small regions and thus replicates partial obstruction by smoke, foliage, or debris, while Shift–Scale–Rotate provides additional geometric diversity in terms of scale and position.

All the geometric augmentations are identically applied to both RGB and simulated thermal channels to preserve pixel-to-pixel alignment between modalities; photometric augmentations, such as color jitter, are applied only to the visible channels in order to prevent corruption of the thermal signal. These augmentations encourage the models to learn viewpoint-invariant, discriminative features leading to performance that is robust against changes in viewpoint and environmental noise and partial visibility of the flame.

IV. PROPOSED METHOD

A. Four-Channel ResNet-34 Architecture

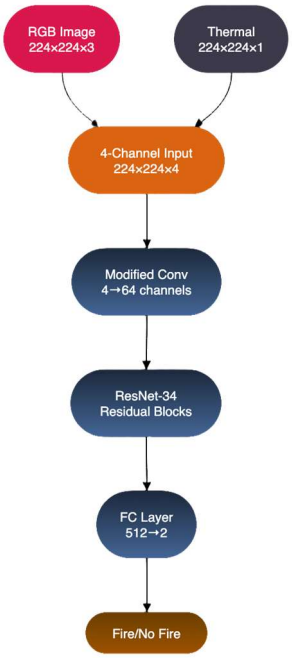


Fig 2. Schematic diagram of the proposed 4-channel ResNet-34 architecture.

The given framework extends the standard ResNet-34 architecture [17] to accept four-channel input tensors, in addition to the common three for both visual and thermal features of fire scenes. Specifically, the extra channel supplies complementary thermal information to color and texture information from RGB channels. The first convolutional layer has therefore been adjusted to accommodate this larger modality size while preserving the pre-trained weight distribution of the base model.

ResNet-34 was utilized due to the residual learning mechanism's ability to pass gradients through extremely deep layers without loss of efficiency in order to improve convergence stability. The feature reusage enabled by the residual connections makes it possible to represent difficult hierarchical features, like the contours of a flame, smoke spreading, and background contrast, without significantly increasing computation.

The entire architecture is composed of four primary steps:

Input Stage: This accepts a four-channel image tensor (R, G, B, T). *Initial Convolution and Pooling:* this consists of a 7×7 convolution, normalization, rectified activation, and max pooling for spatial down-sampling. Residual *Feature Extractors*: made up of four sets of residual blocks that increasingly learn mid- and high-level representations of spatial and radiometric features. *Classification Head:* global average pooling layer followed by a fully connected layer with binary output: fire/no-fire.
As evident in Fig. 2, the extra thermal modality has been incorporated into this setup without compromising on efficiency versus interpretability balance characteristic of

ResNet-34. It is thereby a single multimodal detector that attends to visible as well as simulated heat cues at the same time.

B. Rationale for Selecting ResNet-34

There is a trade-off required for balancing the design of the model for real-time wildfire detection between model accuracy, computational complexity, and inference latency. The research compared several state-of-the-art CNN backbones:
While compound scaling can be used to improve the performance of EfficientNet-B0, it is computationally expensive and therefore unsuitable for execution on edge hardware [15].
Introduction of the multi-scale convolutions in Inception-v3 for increased feature diversity, but its sophisticated architecture creates higher inference times [16].
The MobileNetV3 consumes less memory at the cost of some accuracy, particularly for small or occluded flames.

ResNet-34 offers a very good compromise among these models; it provides competitive accuracy of roughly 95%, moderate parameters of roughly 21 million, and efficient inference, thereby appropriate for UAVs and IoT-based edge devices with the requirements of strict power efficiency and low latency.

C. Merger Strategy

Early data-level fusion is employed in the given system to combine RGB and simulated thermal information prior to feature extraction. This follows previous RGB–thermal work [9]. With this method, the four channels are fused at the input level and learn cross-modal interactions directly from raw data using convolutional layers.

However, it is against feature-level fusion-which would combine the features in intermediate layers-or even decision-level fusion, which puts together predictions following independent processing. Early fusion allows the model to learn spatial–radiometric relationships between modalities:
The RGB channels provide detail in flame geometry, texture, smoke pattern, and contextual background.

The thermal channel introduces pseudo heat-distribution cues, which direct attention to high-intensity ignition locations even under low-light or smoke-obscured situations.
This multimodal fusion serves to increase the robustness of detection in challenging, real-world situations when the visible cues might be incomplete or equivocal.

D. Training Configuration

Model optimization adhered to the traditional practices of supervised learning, with modifications for class imbalance and stable convergence. Weighted cross-entropy loss ensured the learning was evenly balanced between the majority and minority classes. The Adam optimizer ensured adaptive updates with controlled weight decay, and a cyclical learning-rate schedule ensured efficient convergence and alleviated overfitting.

The model was trained on 20 epochs with a mini-batch size of 32. The hyperparameters were also tuned further so that the performance is made stable over the validation set, via grid-based tuning. Random seed initialization also ensured reproducibility, and all experiments were conducted in a GPU environment for faster convergence with the same accuracy.

The setup achieved fast, stable optimisation and high validation accuracy, proving the use of multimodal data can be very well utilized in conventional deep-learning frameworks.

E. Implementation Overview

This modular framework consists of a system to synchronise RGB and synthetic thermal inputs, and the model component to perform four-channel feature extraction and classification. In addition, there are distinct training, validation, and test data streams that provide strict statistical separation of these phases. Its design is conducive to on-the-fly data augmentation and batch processing in an efficient manner, thus ensuring high throughput at training and inference. This combined design thus presents a deployable and scalable framework for real-time wildfire detection through the integration of the established stability of ResNet architectures with increased perceptual ability provided by multimodal fusion.

V.EXPERIMENTAL RESULTS

A. Evaluation Metrics

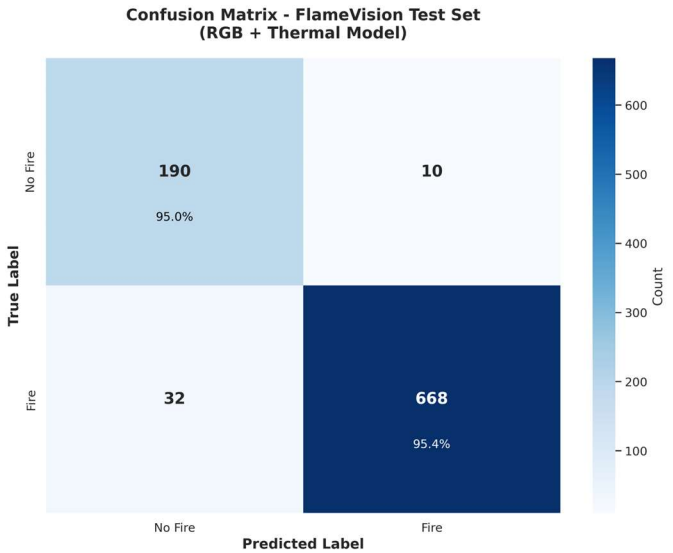


Fig 3. Confusion matrix of fire/no-fire classification results on the test set.

The proposed model's performance was quantitatively evaluated based on the standard metrics of accuracy, precision, recall, and F1-score, which have been defined below:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

$$\text{Precision} = \frac{TP}{TP + FP}, \text{Recall} = \frac{TP}{TP + FN}$$

$$F1 - score = 2 \times \frac{\text{Pr}\, e\, cision \times Recall}{\text{Pr}\, e\, cision + Recall}$$

where TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. Furthermore, confusion matrices were generated to visualize the distribution of misclassified samples and identify weaknesses that are class-specific (see Fig. 3).

B. Performance Comparison

Table I presents the performance comparison between the proposed 4-channel ResNet-34 and the baseline architectures. In turn, the 4-channel network outperforms both RGB-only and EfficientNet-B0 by providing higher accuracy and F1-scores with modest parameter complexity. These results confirm that the effectiveness of simulated thermal fusion has enhanced the fire recognition accuracy.

Model	Accuracy (%)	F1-Score	Params (M)
RGB-only ResNet-34	91.5	0.914	21.8
EfficientNet-B0	94.0	0.939	25.6
Proposed 4-Channel ResNet-34	**95.3**	**0.953**	22.0

Table I: Performance comparison between baseline and proposed architectures.

On average, the inclusion of the simulated thermal channel contributes a 3.8% increase in accuracy relative to the RGB-only baseline and demonstrates that multimodal fusion is very effective even without real thermal sensors.

C. Visual Analysis

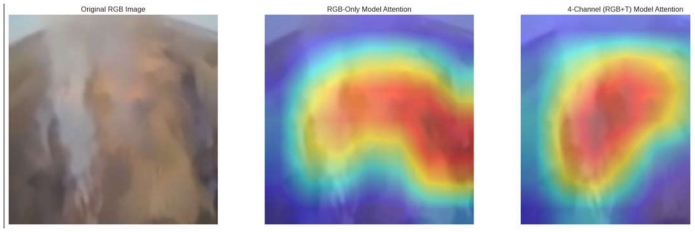


Fig 4. Grad-CAM visualization showing model attention on RGB-only (left) vs. 4-channel (right).

To provide deeper interpretation of network behavior, the current work investigated feature localization by using Grad-CAM visualization. As shown in Fig. 4, the 4-channel ResNet-34 has always demonstrated good activation around flame contours and smoke boundaries, showing robust spatial attention. In contrast, RGB-only models usually misclassify bright regions irrelevant to fires, like sunlight reflections. The addition of simulated thermal information therefore allows the network to discriminate between heat-dominant regions and leads to an improvement in robustness to environmental variability.

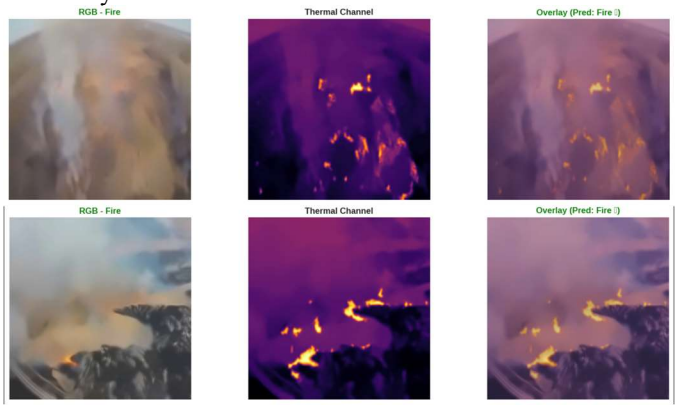


Fig. 5. Qualitative results of the proposed 4-channel ResNet-34 model on test samples.

D. Ablation Study

An ablation analysis was performed to assess the contribution of each input configuration. As shown in Table II, the addition of the simulated thermal channel significantly boosts the performance of classification compared to single-modality RGB input. Replacing the thermal channel with random noise or even raw HSV data resulted in a large drop in performance, confirming the discriminative value of the synthetic thermal modality.

Configuration	Accuracy (%)
RGB only	91.5
RGB + Simulated Thermal	**95.3**
RGB + Random Noise Channel	90.4
RGB + HSV (no fusion)	92.0

Table II: Ablation study of the impact of different channel configurations.

Results from this study confirm that the model's fire-detection capability in difficult conditions, such as when there is smoke or partial occlusion, is enhanced by adding simulated thermal information.

E. Comparison with Real RGB–Thermal Models

The competitiveness of the proposed approach was compared with state-of-the-art real RGB-Thermal models employing physical infrared sensors. CT-Fire [8], using DenseNet and EfficientNet architectures, reported 99.6% accuracy on aerial imagery from dual-sensor data and CP-YOLOv11-MF [9]

reported 96.3% mAP@50 on the RGBT-3M UAV dataset. On the other hand, the proposed 4-channel ResNet-34 reported 95.3% accuracy, utilizing synthetically generated thermal maps, thereby indicating that the simulated thermal modality can approximate real thermal performance at a fraction of the cost. Though models like CT-Fire offer a bit higher accuracy, they require dual-sensor calibration, increased power consumption, and high-end GPU resources for real-time processing. Conversely, the proposed network maintains competitive detection performance with minimal computational overhead and is suitable for UAV-based and edge-deployable wildfire detection systems.

VI.DISCUSSION

A. Synergy of RGB and Thermal Fusion

These results are consistent with existing work in multimodal fusion for visual recognition [8], [9], [10]. The blending of thermal data with visible-spectrum information enhances the performance of the model in detecting fire areas, especially under adverse conditions such as low light, heavy smoke, and partial occlusion. Our suggested 4-channel ResNet-34 could well capture both visual and heat-related signals, taking full advantage of its residual connections in allowing stable gradient flow and effective multimodal feature learning.

Although the thermal modality used in the study is synthetically generated, the simulated thermal maps accurately replicate the spatial and intensity gradients found in real infrared imagery [10]. This proves software-based thermal simulation to be a workable alternative to hardware implementation of thermal imaging. Therefore, the proposed fusion scheme provides a cost-effective solution for comprehensive monitoring of wildfires, where the use of dual-sensor systems would be impractical.

B. Data and Model Generalization

Data augmentation and synthetic thermal generation were performed to better generalize the models for different environmental sets. On unseen wildfire images of the SWIFT validation subset [10], the model attained an accuracy of 93.7%, which was indicative of successful transfer between datasets. This cross-domain resilience speaks to the usability of the proposed system for sim-to-real transfer learning with applicability in deployable environments outside of the training distribution [17].

The findings also underscore that artificial thermal data improves the discriminative power of CNNs through encoding temperature-like features, hence minimizing overfitting to color-based information. This characteristic is especially useful for applications with dynamic backgrounds, atmospheric distortion, or heterogeneous terrain.

C. Computational Efficiency

Computational efficiency was evaluated by testing the performance of the model on an NVIDIA RTX 3060 GPU. The network, with an average inference time of 21.4 milliseconds per image and using 1.9 GB of memory, yielded a throughput of approximately 46 FPS. This is real-time processing capability, appropriate for both UAV-based and embedded monitoring systems.

By contrast, EfficientNet-B0 had a slightly quicker inference time of 19 ms but at the cost of lower detection accuracy. The 4-channel ResNet-34, on the other hand, presents an efficient compromise between computational speed and prediction accuracy and, as such, remains more appropriate for edge computing devices where power and processing resources are scarcer.

D. Limitations

Although the proposed approach gives quite promising results, there are a number of limitations in the suggested methodology. First, the approximation of thermal simulation may not provide the spectral fidelity of actual infrared sensors and hence will limit the performance of such methods in scenes with very faint temperature gradients. Second, the diversity of the data is still rather limited; future research should include real multimodal imagery, such as the RGBT-3M dataset [9], to make the model more realistic and generalizable. Finally, the present framework also deals with a binary classification problem of fire or no fire. Extension to a multi-class detection model with smoke, ember glow, and different levels of fire intensities would yield better situational awareness. Incorporating temporal context from video sequences and using attention mechanisms or vision transformers would enhance the interpretability of the model and its spatiotemporal performance.

VII. CONCLUSION AND FUTURE WORK

This paper presented an architecture of a four-channel ResNet-34 to combine RGB and simulated thermal data for efficient and low-cost wildfire detection. The work proposes a software-based method for multimodal fusion, in which thermal information is synthetically generated from RGB imagery to represent infrared characteristics. In this design, there is no need for a dual-sensor hardware, reducing the overall system complexity and deployment cost.

The experimental results of the FlameVision dataset show that our proposed model achieves a classification accuracy of 95.3%, outperforming RGB-only and EfficientNet-based baselines. It demonstrated that the incorporation of simulated thermal features significantly enhanced the model's performance for identifying fire regions under low-contrast and smoke-obscured conditions. Besides that, the architecture supported real-time inference speeds and low computational costs, which is favorable for deployment on UAVs, surveillance towers, and IoT-based early warning systems. Compared with state-of-the-art multimodal frameworks such as CT-Fire (2024) and CP-YOLOv11-MF (2025), the proposed method achieved similar performances while relying solely on

synthetic thermal generation, underlining its practical utility in resource-constrained environments.

In our future work, we will focus on the use of real thermal imagery sourced from UAVs or satellites, such as the RGBT-3M dataset, for better physical realism. Attention mechanisms such as CBAM or SE blocks can be added to further help refine the weight of multi-modal features and interpretability. The further development of this binary classification framework into a multi-class one would be able to provide smoke, haze, and ember intensity-related categories, thus offering a more comprehensive fire analysis. Future efforts will also be put into its deployment on embedded edge platforms like the NVIDIA Jetson and Raspberry Pi and the development of an end-to-end AI-based early warning system that integrates thermal simulation with real-time camera feeds and analytics on the cloud.

The present work thus constitutes a new and scalable approach for the realization of low-cost, multimodal detection of wildfires, thereby shortening the gap from a research novelty to implementation in an operational field for environmental protection systems.

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